Primitive

1 Calcolare i seguenti integrali indefiniti. Determinare inoltre la primitiva g tale che $g(2) = 0$

1.1) $\displaystyle\int \left(\sqrt{x} + \frac{1}{x}\right) dx$ **1.2)** $\displaystyle\int \left(x^2 - x + \frac{2}{\sqrt[4]{x}}\right) dx$ **1.3)** $\displaystyle\int \frac{\sqrt{x}+2}{x}\, dx$

1.4) $\displaystyle\int (x - \sin x)\, dx$ **1.5)** $\displaystyle\int (x^4 + \sqrt{x} - \sqrt[5]{x-1})\, dx$ **1.6)** $\displaystyle\int (x^{1/9} - x^{1/7} + \cos x)\, dx$

1.7) $\displaystyle\int \frac{3}{1+x^2}\, dx$ **1.8)** $\displaystyle\int (x-1)^3\, dx$ **1.9)** $\displaystyle\int \frac{\sqrt{x} + \sqrt[3]{x}}{\sqrt[4]{x}}\, dx$

1.10) $\displaystyle\int \frac{\sqrt[3]{x} + 2x + 6}{\sqrt{x}}\, dx$ **1.11)** $\displaystyle\int (e^x + 3^x + 5^{-x})\, dx$ **1.12)** $\displaystyle\int \left(e^{x-1} + e^x + e^{x+1}\right) dx$

1.13) $\displaystyle\int \frac{x^5 + 3x^4 + 1}{\sqrt[4]{x^3}}\, dx$ **1.14)** $\displaystyle\int (x^2 + \sin x - \cos x)\, dx$ **1.15)** $\displaystyle\int \left(\sqrt{x-1} + \sqrt{x+1}\right) dx$

2 Per ciascuna delle seguenti funzioni, determinare, se possibile, una primitiva sull'intervallo $I = \mathbb{R}$

2.1) $f(x) = \begin{cases} \sin x & \text{se } x \le 0 \\ x^2 & \text{se } x > 0 \end{cases}$ **2.2)** $f(x) = \begin{cases} \cos x & \text{se } x \le 0 \\ e^x & \text{se } x > 0 \end{cases}$

2.3) $f(x) = \begin{cases} 0 & \text{se } x \le 0 \\ \sqrt{x} & \text{se } x > 0 \end{cases}$ **2.4)** $f(x) = \begin{cases} |x+1| & \text{se } x < 0 \\ |x^2 - 5x + 1| & \text{se } x \ge 0 \end{cases}$

2.5) $f(x) = \begin{cases} x+2 & \text{se } x < -1 \\ x^2 & \text{se } -1 \le x \le 1 \\ \sqrt[4]{x} & \text{se } x > 1 \end{cases}$ **2.6)** $f(x) = \begin{cases} -x^{-1} & \text{se } x \le -1 \\ \sqrt{x+2} & \text{se } x \ge -1 \end{cases}$

2.7) $f(x) = \sqrt{|x|}$ **2.8)** $f(x) = e^{|x+2|}$ **2.9)** $f(x) = \begin{cases} \sin x & \text{se } x < 0 \\ e^{x+1} & \text{se } x \ge 0 \end{cases}$

3 Calcolare i seguenti integrali immediati

3.1) $\displaystyle\int x e^{x^2}\, dx$ **3.2)** $\displaystyle\int x \log(x^2)\, dx$ **3.3)** $\displaystyle\int \sin^4 x \cos x\, dx$ **3.4)** $\displaystyle\int \cos^3 x\, dx$

3.5) $\displaystyle\int \frac{\arctan(\log x)}{x}\, dx$ **3.6)** $\displaystyle\int \frac{1}{x\sqrt{\log x}}\, dx$ **3.7)** $\displaystyle\int \cos x\, \cos(\sin x)\, dx$

3.8) $\displaystyle\int \tan x\, dx$ **3.9)** $\displaystyle\int \frac{\tan x}{\cos x}\, dx$ **3.10)** $\displaystyle\int (4x-2)\sqrt{1+x^2-x}\, dx$

3.11) $\displaystyle\int \frac{x}{1+x^4}\, dx$ **3.12)** $\displaystyle\int \frac{1}{\sqrt{1-4x^2}}\, dx$ **3.13)** $\displaystyle\int \frac{2x}{\sqrt{1-3x^4}}\, dx$

3.14) $\displaystyle\int \sin x \cos x \sqrt[3]{2 + \sin^2 x}\, dx$ **3.15)** $\displaystyle\int \frac{e^{\arctan x}}{1+x^2}\, dx$ **3.16)** $\displaystyle\int \frac{\cos(\log x)}{x} e^{\sin(\log x)}\, dx$

3.17) $\displaystyle\int \frac{\sin x}{1 + \cos^2 x}\, dx$ **3.18)** $\displaystyle\int \frac{\cos x}{\sqrt{1 - \sin^2 x}}\, dx$ **3.19)** $\displaystyle\int \frac{1}{x^2} \arctan\left(\frac{1}{x}\right) dx$

3.20) $\displaystyle\int \frac{1}{\sqrt{(1 - \arcsin^2 x)(1-x^2)}}\, dx$ **3.21)** $\displaystyle\int 2e^x\, e^{e^x}\, dx$

4 Calcolare i seguenti integrali indefiniti, mediante integrazione per parti

4.1) $\int x \sin x \, dx$ **4.2)** $\int x^2 \cos x \, dx$ **4.3)** $\int x \log^2 x \, dx$ **4.4)** $\int \arctan x \, dx$

4.5) $\int x \arctan x \, dx$ **4.6)** $\int (x^2 \cos x - xe^{-2x}) \, dx$ **4.7)** $\int e^x \sin x \, dx$ **4.8)** $\int \arccos x \, dx$

4.9) $\int (x+2) \log x \, dx$ **4.10)** $\int \sqrt{x} \log x \, dx$ **4.11)** $\int e^{2x} \arctan(e^{-x}) \, dx$ **4.12)** $\int \frac{\log x}{x^2} \, dx$

5 Calcolare i seguenti integrali indefiniti di funzioni razionali fratte

5.1) $\int \frac{x}{x+2\pi} \, dx$ **5.2)** $\int \frac{2x+3}{x-4} \, dx$ **5.3)** $f(x) = \int \frac{x-8}{8x-1} \, dx$ **5.4)** $\int \frac{x^2+1}{x+1} \, dx$

5.5) $\int \frac{2x+5}{(x-3)^2} \, dx$ **5.6)** $\int \frac{x}{x^2-6x+5} \, dx$ **5.7)** $\int \frac{x+4}{2x^2-x-6} \, dx$

5.8) $\int \frac{2+x^2}{1+x^2} \, dx$ **5.9)** $\int \frac{x^3+1}{x^2-1} \, dx$ **5.10)** $\int \frac{3x-5}{x^2+4x+5} \, dx$ **5.11)** $\int \frac{1}{x^2+x+1} \, dx$

5.12) $\int \frac{1}{3x^3+x^2-4x} \, dx$ **5.13)** $\int \frac{2x^2+x+1}{x^3+1} \, dx$ **5.14)** $\int \frac{2x+3}{x^3-x} \, dx$

5.15) $\int \frac{x^2+3x+5}{x^3-2x^2+2x-1} \, dx$ **5.16)** $\int \frac{x^3+2x^2-4x-2}{x^3+x^2-x-1} \, dx$ **5.17)** $\int \frac{2x+1}{(x-1)^2(x^2+1)} \, dx$

5.18) $\int \frac{1}{(x^2+1)(x^2+2)} \, dx$ **5.19)** $\int \frac{x^3-2x+1}{(x-2)^2(x^2-7x+12)} \, dx$ **5.20)** $\int \frac{1}{1+x^4} \, dx$

6 Calcolare per sostituzione i seguenti integrali indefiniti, ad esempio tramite la sostituzione suggerita

6.1.a) $\int \frac{e^x}{1+2e^x}$ **6.1.b)** $\int \frac{e^{3x}}{e^{2x}+e^x-1}$ **6.1.c)** $\int \frac{2}{3+4e^x} \, dx$ $(e^x = t)$

6.2.a) $\int \frac{1+\log x}{x \log x} \, dx$ **6.2.b)** $\int \frac{\log^3 x}{x + x \log^2 x} \, dx$ $(\log x = t)$

6.3.a) $\int \frac{1+x^{4/3}}{1+x^{2/3}} \, dx$ **6.3.b)** $\int \frac{\sqrt{x}}{1+\sqrt[3]{x}} \, dx$ $(x^\alpha = t$ con α scelto opportunamente$)$

6.4.a) $\int \frac{1}{\sin x} \, dx$ **6.4.b)** $\int \frac{1}{\sin x + 2\cos x} \, dx$ **6.4.c)** $\int \frac{2+\sin x}{3+\cos x} \, dx$ **6.4.d)** $\int \frac{1}{1+\tan x} \, dx$ $(\tan \frac{x}{2} = t)$

6.5.a) $\int \frac{1}{4-3\cos^2 x + 5\sin^2 x} \, dx$ **6.5.b)** $\int \frac{1+\tan x}{3-4\tan x} \, dx$ $(\tan x = t)$

6.6.a) $\int \sqrt{a^2-x^2} \, dx$ **6.6.b)** $\int \frac{\sqrt{a^2-x^2}}{x} \, dx$ $(x = a \sin t)$

Integrali della forma $\int R\left(x, \sqrt[n]{\frac{ax+b}{cx+d}}\right) dx$, dove R è una funzione razionale fratta in due variabili, possono essere ricondotti a integrali di funzioni razionali fratte mediante la sostituzione $\sqrt[n]{\frac{ax+b}{cx+d}} = t$. Calcolare:

6.7.a) $\int \frac{2}{3+4\sqrt{x}}$ **6.7.b)** $\int \frac{\sqrt{x}+x\sqrt{x}}{1+x} \, dx$ **6.7.c)** $\int \frac{2+\sqrt{x+2}}{3+\sqrt{x+2}} \, dx$ **6.7.d)** $\int \frac{\sqrt[3]{x+1}}{2x+3} \, dx$

6.7.e) $\int \frac{1}{x}\sqrt{\frac{1-x}{1+x}} \, dx$ **6.7.f)** $\int \frac{1+\sqrt{x+1}}{\sqrt{x+1}\,(2-\sqrt{x+1})} \, dx$